

Ahmed Elbossily

CONTACT INFORMATION

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Website: AhmedElbossily.github.io

EDUCATION

Leuphana University Lüneburg
Institute for Production Technology and Systems

Lüneburg, Germany
2024–Present

Ph.D. Candidate

- Advisor: Prof. Dr.-Ing. Benjamin Klusemann
- Dissertation Topic: “GPU-Accelerated Smoothed Particle Hydrodynamics for Simulating Solid-State Manufacturing Processes”

**Arab Academy for Science, Technology, and
Maritime Transport (AASTMT)**

Cairo, Egypt
2014–2018

Master of Science (M.Sc.) in Computational Mechanics

- Grade: Excellent
- Advisor: Prof. Mostafa R. A. Atia

WORK EXPERIENCE

Research Assistant & Ph.D. Candidate

Leuphana University
2024–Present

- Conduct research within an ERC-funded project focused on GPU-accelerated solvers for manufacturing-process simulations.
- Develop and optimize high-performance C++/CUDA algorithms for large-scale particle-based simulations on NVIDIA GPUs.
- Profile and analyze CUDA developed solvers using NVIDIA Nsight Systems and Nsight Compute to identify performance bottlenecks, memory inefficiencies, and kernel-level optimization opportunities.
- Implement and test numerical models on NVIDIA GPUs.
- Supervise graduation projects focused on GPU programming and parallel computing techniques.

Independent Consultant – Udacity

Udacity – Remote
2018–Present

- Deliver mentoring, live online sessions, project reviews, and technical support to improve learner outcomes.
- Support corporate upskilling and accelerator programs, including C++ training for programmers from *Stellantis*.
- Mentor learners in C++, computer vision, machine learning, and artificial intelligence.
- Edit, review, and audit course materials, including the Autonomous Systems curriculum.

Engineering & Technical Leadership Roles

Egypt, Government Organization
2013–2024

Progressed through engineering, R&D, laboratory-management, advanced-systems, team lead, and technical-leadership positions.

- Develop software tools and engineering solutions for technical and operational challenges.
- Implement and optimize GPU-accelerated algorithms, focusing on parallel execution and efficient memory access.
- Profile CUDA applications using NVIDIA Nsight Systems and Nsight Compute.
- Plan and execute complex system tests, with a focus on control-system performance and ECU response.

- Analyze test data, diagnose system-level problems, and recommend technically feasible solutions.
- Led multidisciplinary engineering teams and prepared technical documentation, reports, and process instructions.

TEACHING EXPERIENCE	Leuphana University 2024–Present • Introduction to GPU Programming • GPU Programming for Business Applications • C++ for Mechanical Applications • Mathematics I–II (Exercises)
REFEREED PUBLICATIONS	1. Ahmed Elbossily, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, M. Bambach, and B. Klusemann, “GPU-accelerated meshfree computational framework for modeling the friction surfacing process,” <i>Computational Particle Mechanics</i>, Vol. 12, No. 5, pp. 3721–3745, 2025. https://doi.org/10.1007/s40571-025-01048-2 2. Ahmed Elbossily, Z. Kallien, L. Rath, R. Chafle, M. Afrasiabi, and B. Klusemann, “Experimental–numerical investigation of force-controlled friction extrusion process via feedback-controlled simulation,” <i>Advances in Industrial and Manufacturing Engineering</i>, Vol. 12, 100184, 2026. https://doi.org/10.1016/j.aime.2026.100184 3. Ahmed Elbossily, Z. Kallien, R. Chafle, K. A. Fraser, M. Afrasiabi, and B. Klusemann, “Numerical Investigation of Deposition Efficiency Influencing Factors in the Friction Surfacing Process,” <i>Key Engineering Materials</i>, Vol. 1050, pp. 1–7, April 2026. https://doi.org/10.4028/p-4tzjoz
TECHNICAL SKILLS	Programming: C++, CUDA C/C++, Java, JavaScript, Python Applications: NVIDIA Nsight Systems, Nsight Compute, GLM/C++, Visual Studio, Android Studio, MATLAB, TensorFlow, Inventor, SolidWorks, Abaqus, ParaView Operating Systems: Microsoft Windows and Linux
REFERENCES	Prof. Dr.-Ing. Benjamin Klusemann (Supervisor) Professorship for Materials Mechanics Institute of Production Engineering and Systems Leuphana University Lüneburg E-mail: benjamin.klusemann@leuphana.de